Proving angle relationships

1. The rays $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are opposite rays.
   Determine whether the following are true statements.

   a. $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are the legs of a straight angle.

   b. $m\angle EDF = 180^\circ$

   c. $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are the legs of a right angle.

   d. $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are not the legs of any angle.

   e. $m\angle EDF = 0^\circ$

2. The rays $\overrightarrow{DE}$ and $\overrightarrow{DF}$ are perpendicular to one another.
   Determine whether the following statements are true.

   a. $\angle EDF$ is a right angle.

   b. $m\angle EDF = 180^\circ$

   c. $m\angle EDF = 90^\circ$

   d. $\angle EDF$ is a straight angle.

3. Given that two angles form a linear pair, determine whether the following statements are always true.

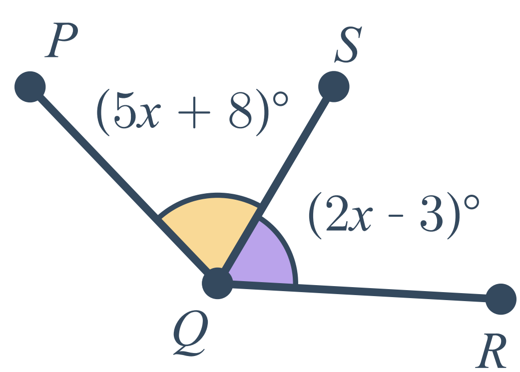
   a. They are complementary.

   b. They are congruent.

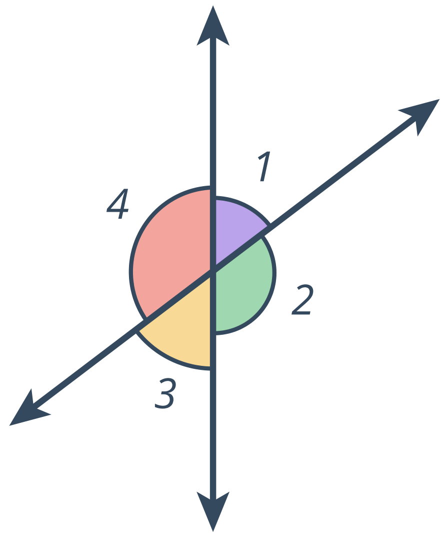
   c. They are adjacent.

   d. They are supplementary.

4. Consider the attached diagram.
   Solve for$x$, given that$m\angle PQR = 145^\circ$.
   Justify each steps using a two column proof.



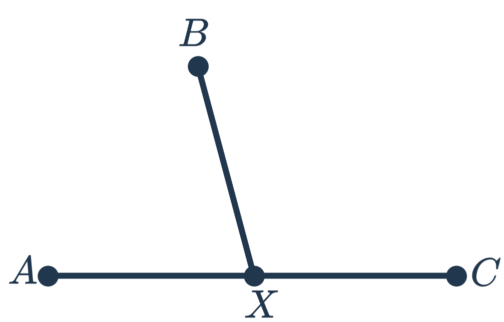
5 Without directly using the vertical angles congruence theorem, construct a two column proof to prove that $\angle 1 \cong \angle 3$.



Additional questions

6 We are given that the two marked angles are congruent.
Construct a two column proof to solve for x.

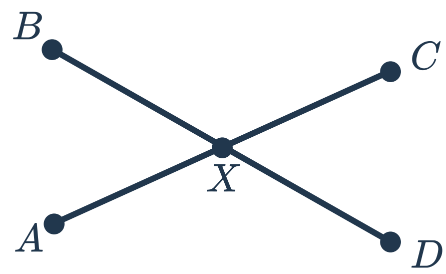
7 Write a two column proof to prove the linear pairs postulate.
Given: $\angle AXB$ and $\angle BXC$ are a linear pair
Prove: $\angle AXB$ and $\angle BXC$ are supplementary



8 Write a two column proof to prove the vertical angles theorem.
Given:

- $\overline{AC}$ is a straight line segment
- $\overline{BD}$ is a straight line segment

Prove: $\angle AXB \cong \angle CXD$



9 Construct a two column proof to prove the congruent complements theorem.
Given:

- $\angle A$ and $\angle C$ are complementary
- $\angle B$ and $\angle C$ are complementary

Prove: $\angle A \cong \angle B$

10 Construct a two column proof for the theorem "all right angles are congruent".
Given: $\angle A$ and $\angle B$ are right angles
Prove: $\angle A \cong \angle B$

11 Write a two column proof for the theorem "if two angles are congruent and supplementary, both are right angles".
Given: $\angle A$ and $\angle B$ are congruent and supplementary
Prove: $\angle A$ and $\angle B$ are right angles